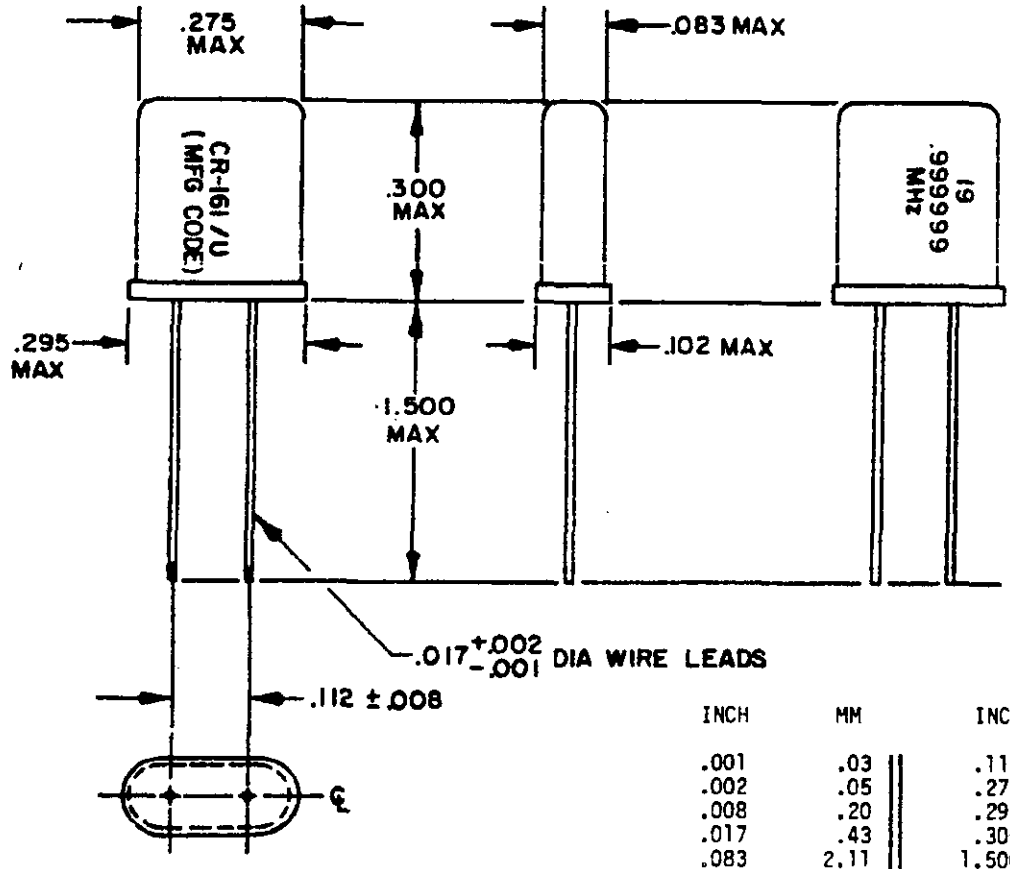
MILITARY SPECIFICATION SHEET

CRYSTAL UNITS, QUARTZ, CR-161/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 16 to 20 MHz; fundamental; noncontrolled; series.



INCH	MM	INCH	MM
.001	.03	.112	2.84
.002	.05	.275	6.99
.008	.20	.295	7.49
.017	.43	.300	7.62
.083	2.11	1.500	38.10
.102	2.59		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type designation, frequency, and code-designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-161/U.

(R) denotes changes

FSC 5955

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TABLE I. *Requirements*.

Requirements	Specified value or condition
Holder	HC-44/U (See MIL-H-10056/31)
Dimensions, marking, and configuration	See figure 1
Frequency range	16 to 20 MHz, inclusive
Frequency tolerance:	
Operating temperature range	±50 ppm
Equivalent resistance	75 ohms maximum
Mode of oscillation	Fundamental
Temperature range:	
Operating	-55°C to +125°C, inclusive
Test set, calibration values, and rated drive level	See table II
Capacitance, shunt	7 pF maximum
Shock (specified pulse): 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 2/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging (100 percent testing):	
Frequency change permitted	±5 ppm

1/ Test condition G.

2/ Method 204 of MIL-STD-202, test condition D and Method 214 of MIL-STD-202, test condition IJ.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mW
TS-683/TSM	16+ to 20	12	--	0.15	2.0 ±0.5

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(B) QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

(B) Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0617-15)

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